# Gravity on the 7+1 spacetime and the coupling of dirac16complex to it

Einstein and Lovelock tensors, generalized frame matrices, the canonical spin connection with a vanishing total covariant derivative, the covariant Dirac equation of dirac16complex, and the primordial, modern and far-future epochs — sections 11–14 of the notebook 7+1.nb

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**Abstract**

This document puts the 16-component field dirac16complex of the 7+1 project into a gravitational field. The spacetime has ONE time coordinate $x_0$ and SEVEN space coordinates $x_1, \ldots, x_7$; the four auxiliary space-like coordinates $x_4, \ldots, x_7$ may expand or contract at their own rate. It defines and checks the curvature of a general metric, the Lovelock tensors of 7+1 spacetime (Einstein, Gauss–Bonnet, cubic; the quartic one vanishes identically in eight dimensions) and the Einstein–Lovelock field equations; the generalized frame matrices $\Gamma_\mu(X) = e^A{}_\mu(X)\,\Gamma_A$ built from the real 16×16 Dirac matrices; the canonical spin connection, for which the total covariant derivative of every frame matrix vanishes; the covariant Dirac equation, in which that spin connection couples gravity to dirac16complex; and the three cosmological epochs — primordial (near $x_0 = 0$), modern, far future — with the creation of dirac16complex pairs in each. The ideas follow the repository Dirac_claude, rebuilt for 7+1: Dirac_claude uses a 4+4 spacetime with four time-like coordinates and a warped metric, none of which is used here. Sections 11–14 of 7+1.nb contain 100 checks (100 True); the independent Python program tools/ref_gravity.py contains 101 (101 True).

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# 1. Results at a glance

**Spacetime.** Coordinates $X = (x_0, x_1, \ldots, x_7)$ with $x_0$ the time (the direction of the octonion unit $e_0 = 1$), $x_1, x_2, x_3$ ordinary space and $x_4, \ldots, x_7$ auxiliary space: one time-like and seven space-like dimensions, which xAct declares, as the reference notebook Einstein-Lovelock-7+1-Nash.nb does, with the signature list $\{7, 1, 0\}$ (seven space-like, one time-like, none degenerate). The frame metric — simply the squares of the octonion basis, $e_0^2 = +1$, $e_i^2 = -1$ (section 2) — and the eight real Dirac matrices are

$$\eta = \mathrm{diag}(+1, -1, \ldots, -1), \qquad \Gamma_0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \quad \Gamma_i = \begin{pmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{pmatrix}, \qquad \Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\eta_{AB}\,1_{16},$$

with the real antisymmetric 8×8 octonion matrices $\tau_i^O$ ($i = 1, \ldots, 7$, $(\tau_i^O)^2 = -1_8$, mutually anticommuting).

**Lovelock tensors** of order $k$, with the generalized Kronecker delta $\delta^{\mu_1\cdots\mu_n}_{\nu_1\cdots\nu_n} = \det[\delta^{\mu_i}_{\nu_j}]$:

$$E^{(k)\mu}{}_{\nu} = -\frac{1}{2^{k+1}}\,\delta^{\mu\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\nu\gamma_1\delta_1\cdots\gamma_k\delta_k}\,R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k},$$
$$E^{(0)} = -\tfrac{1}{2}\,\delta, \quad E^{(1)} = G \text{ (Einstein)}, \quad E^{(4)} \equiv 0 \text{ in } D = 8.$$

**Einstein–Lovelock field equations** ($\alpha_0 = 2\Lambda$, $\alpha_1 = 1$, $\kappa = 8\pi G_8$):

$$\sum_{k=0}^{3} \alpha_k\, E^{(k)\mu}{}_{\nu} = \kappa\, T^{\mu}{}_{\nu}.$$

**Generalized frame matrices, canonical spin connection, total covariant derivative:**

$$\Gamma_\mu(X) = e^A{}_\mu(X)\,\Gamma_A, \qquad \Gamma_\mu\Gamma_\nu + \Gamma_\nu\Gamma_\mu = 2g_{\mu\nu}(X)\,1_{16},$$
$$\omega_\mu{}^A{}_B = e^A{}_\nu\big(\partial_\mu e_B{}^\nu + \{{}^{\nu}_{\mu\lambda}\}\,e_B{}^\lambda\big), \qquad \Omega_\mu = \tfrac{1}{4}\,\omega_{\mu AB}\,\Gamma^A\Gamma^B,$$
$$\nabla_\mu\Gamma_\nu = \partial_\mu\Gamma_\nu - \{{}^{\lambda}_{\mu\nu}\}\Gamma_\lambda + [\Omega_\mu, \Gamma_\nu] = 0.$$

**Covariant Dirac equation of dirac16complex** ($D_\mu = \partial_\mu + \Omega_\mu$, $\bar\psi = \psi^\dagger\Gamma_0$):

$$\big(i\,\Gamma^\mu(X)\,D_\mu - m\big)\psi = 0, \qquad (\Gamma^\mu D_\mu)^2 = g^{\mu\nu}\nabla_\mu D_\nu - \tfrac{R}{4}\,1_{16}.$$

**Three epochs** of the cosmological metric $ds^2 = dx_0^2 - a^2(dx_1^2 + dx_2^2 + dx_3^2) - b^2(dx_4^2 + \cdots + dx_7^2)$: primordial Kasner solutions near $x_0 = 0$ (Einstein: $a = x_0^p$, $b = x_0^q$ with $p = (1 \pm 2\sqrt{2})/7$, $q = (2 \mp 3\sqrt{2})/14$; Gauss–Bonnet and cubic Lovelock exponents below), a modern epoch with a static auxiliary space ($b = b_0$, ordinary Friedmann cosmology with $G_4 = G_8/V_\perp$), and de Sitter far futures, in which dirac16complex pairs are created with the Fermi–Dirac distribution at the Gibbons–Hawking temperature, $|\beta|^2 = 1/(e^{2\pi M/H} + 1)$.

## 2. Conventions

- Units: $\hbar = c = 1$. A prime is $d/dx_0$.
- Indices: Greek $\mu, \nu, \lambda, \rho, \sigma, \alpha, \beta, \gamma, \delta \in \{0, \ldots, 7\}$ are coordinate indices; Latin capitals $A, B, C \in \{0, \ldots, 7\}$ are frame indices, raised and lowered with $\eta = \mathrm{diag}(+1, -1, \ldots, -1)$, the squares of the octonion basis ($\eta_{AB} = \mathrm{Re}(e_A e_B)$); repeated indices are summed. In Mathematica all lists are 1-based: $g_{\mu\nu}$ is `g[[μ+1, ν+1]]`.
- Signature: one time-like dimension ($x_0$, with $g_{00} > 0$) and seven space-like dimensions ($x_1, \ldots, x_7$, negative squares); in xAct the signature $\{7, 1, 0\}$ (seven space-like, one time-like, none degenerate), as in the reference notebook.
- Christoffel symbols are written $\{^{\lambda}_{\mu\nu}\}$ — never $\Gamma$, which in this project always denotes a Dirac matrix:

$$\{^{\lambda}_{\mu\nu}\} = \tfrac{1}{2}\, g^{\lambda\rho}\big(\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\big).$$

Curvature (the convention of Landau and Lifshitz):

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \{^{\rho}_{\sigma\nu}\} - \partial_\nu \{^{\rho}_{\sigma\mu}\} + \{^{\rho}_{\mu\lambda}\}\{^{\lambda}_{\sigma\nu}\} - \{^{\rho}_{\nu\lambda}\}\{^{\lambda}_{\sigma\mu}\},$$
$$R_{\sigma\nu} = R^{\rho}{}_{\sigma\rho\nu}, \qquad R = g^{\sigma\nu} R_{\sigma\nu}, \qquad G^{\mu}{}_{\nu} = R^{\mu}{}_{\nu} - \tfrac{1}{2}\delta^{\mu}_{\nu} R, \qquad R^{\gamma\delta}{}_{\alpha\beta} = g^{\delta\sigma} R^{\gamma}{}_{\sigma\alpha\beta}.$$

With these conventions and the signs of $\eta$ ($g_{00} > 0$) the field equations of general relativity read $G^{\mu}{}_{\nu} - \Lambda\delta^{\mu}_{\nu} = \kappa T^{\mu}{}_{\nu}$ with $T^0{}_0 > 0$ the energy density, and a de Sitter space has $R < 0$.

**The signature: $\{7, 1, 0\}$ and $\mathrm{diag}(+1, -1, \ldots, -1)$.** The 7+1 spacetime has one time-like dimension ($x_0$, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions ($x_1, x_2, x_3$ ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7, 1, 0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and $-1$ of $\eta_{71} = \mathrm{diag}(-1, +1, \ldots, +1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like = $+1$, time-like = $-1$). (ii) The matrix $\eta = -\eta_{71} = \mathrm{diag}(+1, -1, \ldots, -1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1, \ldots, 7),$$
$$\mathrm{Re}(x\, x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

$\eta$ gives the signs of the Clifford relation $\Gamma_A \Gamma_B + \Gamma_B \Gamma_A = 2\,\eta_{AB}\, 1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to $-1_{16}$, exactly like the octonion units (in the notation $Cl(p, q)$, with $p$ generators squaring to $+1$ and $q$ to $-1$, the algebra $Cl(1, 7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det \eta = \det \eta_{71} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7, 1, 0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\, x) = \eta_{AB} x^A x^B$.

## 3. Curvature of a 7+1 metric (section 11 of 7+1.nb)

The notebook function `curvature[g, simp]` returns, for an 8×8 metric in the coordinates $X$, the inverse metric, the Christoffel symbols, $R^{\rho}{}_{\sigma\mu\nu}$, $R^{\gamma\delta}{}_{\alpha\beta}$, the Ricci tensor, $R$ and $G^{\mu}{}_{\nu}$. Three test metrics are used throughout:

1. the **cosmological metric** $g = \mathrm{diag}(1, -a^2, -a^2, -a^2, -b^2, -b^2, -b^2, -b^2)$ with arbitrary $a(x_0)$ (`a3[x0]`) and $b(x_0)$ (`b4[x0]`);
2. an **inhomogeneous metric** $g = \mathrm{diag}(n^2, -a^2, -a^2, -a^2, -b^2, -b^2, -b^2, -b^2)$ in which $n, a, b$ are explicit polynomials of BOTH $x_0$ and $x_4$: $n = 1 + x_0 x_4/5 + x_0^2/11$, $a = 1 + x_0/3 + x_4^2/7$, $b = 2 - x_0 x_4/4 + x_0^2/9 + x_4/13$ (`nP`, `aP`, `bP`); its identities are checked exactly (rational numbers) at the point $x_0 = 1/2$, $x_4 = 1/3$;

3. a **plane gravitational wave** (pp-wave) $ds^2 = dx_0^2 - dx_1^2 - \cdots - dx_7^2 + F(x_0 - x_1, x_2, x_3)\,(dx_0 - dx_1)^2$ with an arbitrary profile $F$ (`fPP`), which is not diagonal.

For the cosmological metric, with the Hubble rates $H_a = a'/a$ and $H_b = b'/b$, the Riemann tensor in the orthonormal frame is diagonal on index pairs; its only components are the sectional curvatures $K_{AB} = R^{AB}{}_{AB}$ (no sum), with $i, j \in \{1, 2, 3\}$, $k, l \in \{4, \ldots, 7\}$:

$$K_{0i} = -(H_a' + H_a^2), \quad K_{0k} = -(H_b' + H_b^2), \quad K_{ij} = -H_a^2, \quad K_{kl} = -H_b^2, \quad K_{ik} = -H_a H_b.$$

The Einstein tensor and the scalar curvature are

$$\begin{aligned} G^0{}_0 &= 3H_a^2 + 12H_aH_b + 6H_b^2, \\ G^1{}_1 = G^2{}_2 = G^3{}_3 &= 2H_a' + 3H_a^2 + 4H_b' + 10H_b^2 + 8H_aH_b, \\ G^4{}_4 = \cdots = G^7{}_7 &= 3H_a' + 6H_a^2 + 3H_b' + 6H_b^2 + 9H_aH_b, \\ R &= -2\big(3H_a' + 6H_a^2 + 4H_b' + 10H_b^2 + 12H_aH_b\big). \end{aligned}$$

For the inhomogeneous metric the notebook checks, exactly at $x_0 = 1/2$, $x_4 = 1/3$, the symmetries of the Riemann tensor ($R_{\rho\sigma\mu\nu} = -R_{\sigma\rho\mu\nu} = -R_{\rho\sigma\nu\mu} = R_{\mu\nu\rho\sigma}$ with $R_{\rho\sigma\mu\nu} = g_{\rho\lambda}R^{\lambda}{}_{\sigma\mu\nu}$), the first Bianchi identity $R^{\rho}{}_{\sigma\mu\nu} + R^{\rho}{}_{\mu\nu\sigma} + R^{\rho}{}_{\nu\sigma\mu} = 0$, and the contracted Bianchi identity

$$\nabla_\mu G^\mu{}_\nu = \partial_\mu G^\mu{}_\nu + \{{}^{\mu}_{\mu\lambda}\} G^\lambda{}_\nu - \{{}^{\lambda}_{\mu\nu}\} G^\mu{}_\lambda = 0.$$

(the contracted Bianchi identity is also checked for the cosmological metric with arbitrary $a(x_0)$, $b(x_0)$). A fourth metric enters in section 5.4: the static metric $ds^2 = f(r)\,dx_0^2 - h(r)\,dr^2 - r^2(dx_2^2 + \cdots + dx_7^2)$ with $r = x_1$ and arbitrary positive $f$, $h$.

# 4. The Lovelock tensors of 7+1 spacetime (section 12)

## 4.1 Lovelock's theorem and the definitions

In $n$ dimensions the most general tensor $A^{lh}$ that is symmetric, divergence-free and built from the metric and its first and second derivatives is (D. Lovelock and H. Rund, *Tensors, Differential Forms, and Variational Principles*, Dover; Remark 2 after their Eq. (4.38))

$$A^{lh} = \sqrt{|g|}\sum_{k=1}^{m-1} \alpha_{(k)}\, g^{jl}\, \delta^{hh_1\cdots h_{2k}}_{jj_1\cdots j_{2k}}\, R^{j_1j_2}{}_{h_1h_2}\cdots R^{j_{2k-1}j_{2k}}{}_{h_{2k-1}h_{2k}} + \lambda\sqrt{|g|}\, g^{lh}, \qquad m = \begin{cases} n/2, & n \text{ even}, \\ (n+1)/2, & n \text{ odd}, \end{cases}$$

with arbitrary constants $\alpha_{(k)}$ and $\lambda$. For $n = 8$, $m - 1 = 8/2 - 1 = 3$: there are exactly three curvature terms. The generalized Kronecker delta is the determinant of the matrix of ordinary Kronecker deltas,

$$\delta^{\mu_1\cdots\mu_n}_{\nu_1\cdots\nu_n} = \det\left[\delta^{\mu_i}_{\nu_j}\right]_{i,j=1}^{n}, \qquad \delta^{\mu_1\cdots\mu_p\rho}_{\nu_1\cdots\nu_p\rho} = (8-p)\,\delta^{\mu_1\cdots\mu_p}_{\nu_1\cdots\nu_p} \quad (D = 8),$$

it is $+1$ ($-1$) when the lower indices are an even (odd) permutation of the upper ones, all different, and 0 otherwise; it vanishes when $n > 8$. With the normalization of the reference notebook Einstein-Lovelock-7+1-Nash.nb (P. L. Nash, GPL), which this document follows, the Lovelock tensors (`Lovelock1`, `Lovelock2`, `Lovelock3`) and densities are

$$\begin{aligned} \text{Lovelock}_k \equiv E^{(k)\mu}{}_\nu &= -\frac{1}{2^{k+1}}\,\delta^{\mu\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\nu\gamma_1\delta_1\cdots\gamma_k\delta_k}\, R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k}, \\ L_k &= \frac{1}{2^k}\,\delta^{\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\gamma_1\delta_1\cdots\gamma_k\delta_k}\, R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k}. \end{aligned}$$

Special cases: $L_0 = 1$, $E^{(0)\mu}{}_\nu = -\frac{1}{2}\delta^\mu_\nu$; $L_1 = R$, $E^{(1)} = G$; $L_2 = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ (Gauss–Bonnet) with $E^{(2)}$ the Lanczos tensor

$$E^{(2)}_{\mu\nu} = 2\big(R\,R_{\mu\nu} - 2R_{\mu\alpha}R^\alpha{}_\nu - 2R^{\alpha\beta}R_{\mu\alpha\nu\beta} + R_\mu{}^{\alpha\beta\gamma}R_{\nu\alpha\beta\gamma}\big) - \tfrac{1}{2}g_{\mu\nu}L_2;$$

$L_3$, $E^{(3)}$ are cubic in the curvature (section 4.3 gives $E^{(3)}$ explicitly). Because $E^{(k)}$ contains $2k+1$ different upper indices, $E^{(4)} \equiv 0$ in eight dimensions, although $L_4$ (the Euler density) is not zero; $\sqrt{|g|}L_4$ is a total derivative. Every $E^{(k)}$ is symmetric with both indices down, divergence-free for every metric, and has the trace

$$E^{(k)\mu}{}_\mu = -\frac{8-2k}{2}\,L_k.$$

## 4.2 Action and field equations

The reference notebook writes the Einstein–Lovelock VACUUM field equations as

$$0 = -\Lambda\,\delta^{\mu}_{\nu} + H^{-2}\,w_1\,\text{Lovelock}_1 + H^{-4}\,w_2\,\text{Lovelock}_2 + H^{-6}\,w_3\,\text{Lovelock}_3,$$

with pure numbers $w_1, w_2, w_3, \Lambda$ and a fundamental inverse length $H$ (it exists because the Lovelock tensors of different orders have different dimensions, length$^{-2k}$). With matter, and with $\delta(\sqrt{|g|}L_k) = \sqrt{|g|}\,E^{(k)}_{\mu\nu}\,\delta g^{\mu\nu}$ (up to a total derivative) and $\delta S_{\text{matter}} = \frac{1}{2}\int \sqrt{|g|}\,T_{\mu\nu}\,\delta g^{\mu\nu}\,d^8X$, the action

$$S = -\frac{1}{2\kappa}\int d^8X\,\sqrt{|g|}\,\sum_{k=0}^{3}\alpha_k L_k + S_{\text{matter}}, \qquad \alpha_0 = 2\bar{\Lambda},\quad \alpha_1 = 1,$$

gives the Einstein–Lovelock field equations

$$\sum_{k=0}^{3}\alpha_k\,E^{(k)\mu}{}_{\nu} = \kappa\,T^{\mu}{}_{\nu} \qquad\Longleftrightarrow\qquad G^{\mu}{}_{\nu} + \alpha_2 E^{(2)\mu}{}_{\nu} + \alpha_3 E^{(3)\mu}{}_{\nu} - \bar{\Lambda}\,\delta^{\mu}_{\nu} = \kappa\,T^{\mu}{}_{\nu},$$

with $\kappa = 8\pi G_8$ ($G_8$ the Newton constant of eight dimensions), the Gauss–Bonnet and cubic couplings $\alpha_2$, $\alpha_3$ (dimensions length$^2$ and length$^4$) and the cosmological constant $\bar{\Lambda}$ (dimension length$^{-2}$; it is written $\Lambda$ in sections 7 and following, where no pure number $\Lambda$ appears). $\alpha_4$ does not appear: its tensor vanishes. For $T = 0$ the two forms are the same equation (divide the first by $w_1/H^2$):

$$\alpha_2 = \frac{w_2}{w_1 H^2}, \qquad \alpha_3 = \frac{w_3}{w_1 H^4}, \qquad \bar{\Lambda} = \frac{\Lambda H^2}{w_1}.$$

## 4.3 The Lovelock tensors with xAct, as in the reference notebook

Section 12 of 7+1.nb computes the Lovelock tensors exactly as the reference notebook does, with the tensor packages xAct (xPerm, xTensor, xCoba, xTras; J. M. Martín-García et al., GPL; versions xPerm 1.2.3, xTensor 1.2.0, xCoba 0.8.6, xTras 1.4.2). The reference notebook also contains a "Model metric" which its author has marked FAILED, DO NOT USE (it never worked for the 7+1 signature); it is not used anywhere in this project. The set-up (abstract indices $a$, $b$, $c$, $d$, $f$, $i$, $j$, $k$, $l$, $m$, $n$, $p$, $q$, $r$, $s$, $y$; a metric `g71` with Levi-Civita derivative `CD`; a chart `chartM8` with the coordinate scalars `x0[]`, …, `x7[]`):

```
Needs["xAct`xCoba`"]; Needs["xAct`xTras`"];
$DefInfoQ = False; $CVVerbose = False;
SetOptions[ToCanonical, UseMetricOnVBundle -> None, Verbose -> False];
DefManifold[M8, 8, {a, b, c, d, f, i, j, k, l, m, n, p, q, r, s, y}];
DefMetric[{7, 1, 0}, g71[-a, -b], CD, {";", "D"}, WeightedWithBasis -> AIndex];
DefChart[chartM8, M8, {0, 1, 2, 3, 4, 5, 6, 7}, {x0[], x1[], x2[], x3[], x4[], x5[], x6[],
    ...x7[]},
  ExtendedCoordinateDerivatives -> True, FormatBasis -> {"Partials", "Differentials"}];
MyArrayComponents[expr_] := expr // ToBasis[chartM8] // ComponentArray // ToValues // ToValues
    ...// Simplify;
kδ[lower_, upper_] /; Length[lower] == Length[upper] := Det[Outer[delta, lower, upper]];
```

**The signature.** `DefMetric[{7, 1, 0}, …]` is the declaration of the reference notebook, taken unchanged: in xAct the list $\{7,1,0\}$ declares seven space-like dimensions $(x_1,\ldots,x_7)$, one time-like dimension $(x_0)$ and no degenerate one — the signature $(7,1)$ of $\eta_{71} = \text{diag}(-1,+1,\ldots,+1)$ in the sign convention of 4+4.nb (space-like $= +1$). It is not to be confused with the matrix $\eta = \eta_{17} = -\eta_{71} = \text{diag}(+1,-1,\ldots,-1)$, which is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units (space), i.e. $\eta_{AB} = \text{Re}(e_A e_B)$ and $\text{Re}(xx) = x_0^2 - x_1^2 - \cdots - x_7^2$ for $x = x^A e_A$. The metrics of this document are given to xAct with the signs of $\eta$ ($g_{00} > 0$). The declared signature fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, the same in both conventions (a metric with one time-like and seven space-like directions has a negative determinant either way); the Lovelock tensors follow from the components of the metric, and they agree exactly with the independent pair-set computation of section 4.5. The notebook checks `SignatureOfMetric[g71]` $= \{7,1,0\}$, `SignDetOfMetric[g71]` $= -1$,

$\eta_{AB} = \mathrm{Re}(e_A e_B)$ and the quadratic form $\mathrm{Re}(xx) = x \cdot \eta \cdot x$ of the octonions. (An earlier version of 7+1.nb declared $\{1, 7, 0\}$, reading the list as the signs of $\eta$; it was corrected on 27 October 2026. Both declarations give the same Lovelock tensors.)

While the packages load, their banners and the warnings that xTensor cannot attach rules to `Commutator` and `Anticommutator` (protected built-in symbols of Wolfram 15, not used here) are suppressed. `kδ[lower, upper]` is the generalized Kronecker delta $\delta^{\text{upper}}_{\text{lower}}$ with abstract indices; the Lovelock tensors are then typed exactly as in the reference notebook,

```
Lovelock1 = -(1/4) RiemannCD[-c,-d,f,i] kδ[{-a,-f,-i},{b,c,d}] // Simplification
Lovelock2 = -(1/8) RiemannCD[-c,-d,f,i] RiemannCD[-j,-k,l,m] kδ[{-a,-f,-i,-l,-m},{b,c,d,j,k}]
    ...// Simplification
Lovelock3 = -(1/16) RiemannCD[-c,-d,f,i] RiemannCD[-j,-k,l,m] RiemannCD[-n,-p,q,r]
              kδ[{-a,-f,-i,-l,-m,-q,-r},{b,c,d,j,k,n,p}] // Simplification
```

i.e. $\text{Lovelock}_k = E^{(k)b}{}_a$ with the free indices $a$ (down) and $b$ (up) and $R_{cd}{}^{fi}$ = `RiemannCD[-c,-d,f,i]`. Simplification (xTras) reduces them to curvature monomials; with $R_{ab}{}^{cd}$ the Riemann tensor, $R_a{}^b$ the Ricci tensor and $R$ the scalar curvature the notebook obtains

$$\text{Lovelock}_1 = -\frac{\delta_a{}^b R}{2} + R_a{}^b,$$

$$\begin{aligned}\text{Lovelock}_2 = &-\frac{\delta_a{}^b R^2}{2} + 2\delta_a{}^b R^j{}_k R_j{}^k \\ &- \frac{\delta_a{}^b R_{jk}{}^{lm} R^{jk}{}_{lm}}{2} + 2R R_a{}^b \\ &- 4R_a{}^j R^b{}_j - 4R^j{}_k R_{aj}{}^{bk} \\ &+ 2R_{aj}{}^{kl} R^{bj}{}_{kl},\end{aligned}$$

$$\begin{aligned}\text{Lovelock}_3 = &-\frac{\delta_a{}^b R^3}{2} + 6\delta_a{}^b R R^j{}_k R_j{}^k \\ &- \frac{3\delta_a{}^b R R_{jk}{}^{lm} R^{jk}{}_{lm}}{2} \\ &- 8\delta_a{}^b R_j{}^l R^j{}_k R^k{}_l \\ &- 12\delta_a{}^b R^j{}_k R^l{}_m R_{jl}{}^{km} \\ &+ 12\delta_a{}^b R^j{}_k R_{jl}{}^{mn} R^{kl}{}_{mn} \\ &+ 4\delta_a{}^b R^{jk}{}_{lm} R_{jn}{}^{lp} R_{kp}{}^{mn} \\ &- \delta_a{}^b R^{jk}{}_{lm} R_{jk}{}^{np} R^{lm}{}_{np} + 3R^2 R_a{}^b \\ &- 12R R_a{}^j R^b{}_j - 12R R^j{}_k R_{aj}{}^{bk} \\ &+ 6R R_{aj}{}^{kl} R^{bj}{}_{kl} + 24R_j{}^l R^j{}_k R_{al}{}^{bk} \\ &+ 24R_a{}^j R^b{}_k R_j{}^k + 24R_a{}^j R^k{}_l R^{bl}{}_{jk} \\ &- 12R_a{}^j R^{bk}{}_{lm} R_{jk}{}^{lm} - 12R^j{}_k R_a{}^b R_j{}^k \\ &- 12R^j{}_k R_{aj}{}^{lm} R^{bk}{}_{lm} \\ &- 24R^j{}_k R_{al}{}^{km} R^{bl}{}_{jm} \\ &+ 24R^j{}_k R_{al}{}^{bm} R_{jm}{}^{kl} \\ &+ 3R_a{}^b R_{jk}{}^{lm} R^{jk}{}_{lm} + 24R^k{}_l R^b{}_j R_{ak}{}^{jl} \\ &- 12R^b{}_j R_{ak}{}^{lm} R^{jk}{}_{lm} \\ &- 24R_{aj}{}^{kl} R^{bm}{}_{kn} R^{jn}{}_{lm} \\ &+ 6R_{aj}{}^{kl} R^{bj}{}_{mn} R_{kl}{}^{mn} \\ &- 12R_{kl}{}^{mn} R^{jl}{}_{mn} R_{aj}{}^{bk}.\end{aligned}$$

Two checks are made with abstract indices (ToCanonical): $\text{Lovelock}_1 - (R_a{}^b - \frac{1}{2}\delta_a{}^b R) = 0$, and the trace $\text{Lovelock}_{2a}{}^a = -2L_2$ with $L_2 = R^2 - 4R_{cd}R^{cd} + R_{cdfi}R^{cdfi}$. The reference notebook sets the option

`UseMetricOnVBundle -> None` of ToCanonical (indices are not moved up and down); for the trace identity the metric must identify the dummy pairs $R_{cd}R^{cd}$ and $R^c{}_dR_c{}^d$, so that one call uses `UseMetricOnVBundle -> All`.

## 4.4 Components with xAct: CD = LC[g]

For a metric given as an 8×8 matrix of components with lower indices the reference notebook makes it a CTensor in the basis of `chartM8`, lets MetricCompute compute and simplify its curvature, and sets `CD = LC[g]` ("xAct magic": the abstract covariant derivative becomes the Levi-Civita connection of the component metric). After `RiemannCD = Riemann[CD]`, `RicciCD = Ricci[CD]`, `RicciScalarCD = RicciScalar[CD]` and `EinsteinCD = Einstein[CD]` every abstract expression evaluates to components. The notebook function `xActLovelock[gm]` does exactly these steps:

```
xActLovelock[gm_] := Module[{res},
   g71 = CTensor[gm /. toXc, {-chartM8, -chartM8}];
   MetricCompute[g71, chartM8, All, Parallelize -> False, Verbose -> False];
   CD = LC[g71];
   RiemannCD = Riemann[CD]; RicciCD = Ricci[CD]; RicciScalarCD = RicciScalar[CD]; EinsteinCD =
    ...Einstein[CD];
   res = Transpose[asMat8[MyArrayComponents[#]]] & /@ {Lovelock1, Lovelock2, Lovelock3,
    ...EinsteinCD[-a, b]};
   res /. fromXc]
```

Here `toXc` replaces $x_\mu$ by the coordinate scalar `xμ[]` and `fromXc` undoes it, `asMat8` turns a vanishing result into the 8×8 zero matrix, and the Transpose makes row $\mu$, column $\nu$ hold $E^{(k)\mu}{}_\nu$ (the array of MyArrayComponents has the lower index $a$ first; the pp-wave of section 4.7, whose $E^{(1)\mu}{}_\nu$ is not a symmetric matrix, fixes this order). For the cosmological metric `xActLovelock` gives $\text{Lovelock}_1$ = `EinsteinCD` = the Einstein tensor of section 3, and the closed forms of section 4.6.

## 4.5 An independent computation: the pair-set algorithm

The same tensors are also computed without xAct. A direct evaluation of the generalized delta would run over $8^{4k+2}$ index combinations; the notebook functions `pairSets`, `orderedSeqs`, `lovelockE`, `lovelockL` (and tools/lovelock71.py in Python) use the antisymmetries instead. Let $\mathcal{U}$ run over the SETS of $k$ disjoint index pairs $(\alpha<\beta)$ not containing $\mu$, and $\mathcal{V}$ over the ORDERED sequences of $k$ disjoint pairs $(\gamma<\delta)$ that exhaust $(\mathcal{U}\cup\{\mu\})\setminus\{\nu\}$. Each pair contributes a factor 2 (both orders of a pair give the same term), the $k!$ orderings of the pairs of $\mathcal{U}$ give the same sum, and

$$E^{(k)\mu}{}_\nu = -2^{k-1}k!\sum_{\mathcal{U}}\sum_{\mathcal{V}}\operatorname{sgn}(\mu\,\mathcal{U}\to\nu\,\mathcal{V})\prod_{i=1}^{k}R^{\mathcal{V}_i}{}_{\mathcal{U}_i}, \qquad L_k = 2^k k!\sum_{\mathcal{U}}\sum_{\mathcal{V}}\operatorname{sgn}(\mathcal{U}\to\mathcal{V})\prod_{i=1}^{k}R^{\mathcal{V}_i}{}_{\mathcal{U}_i},$$

where sgn is the sign of the permutation that turns the upper index sequence into the lower one (in Mathematica `Signature[up] Signature[dn]`). For $k=3$ and fixed $\mu,\nu$ there are 105 upper pair-sets and 90 ordered lower sequences. The notebook compares $E^{(1)}$ with the brute-force contraction $-\frac{1}{4}\delta^{\mu\alpha\beta}_{\nu\gamma\delta}R^{\gamma\delta}{}_{\alpha\beta}$, $E^{(2)}$ with the Lanczos tensor, and $E^{(1)},E^{(2)},E^{(3)}$ of the cosmological metric and of the pp-wave with xAct: the two computations agree exactly. The pair-set sum also works on a table of exact rational numbers (section 4.7).

## 4.6 Closed forms for the cosmological metric

For the cosmological metric every $E^{(k)}$ is diagonal; the entry $E^{(k)\mu}{}_\mu$ (no sum) is $-\frac{1}{2}$ times the density $L_k$ restricted to the index pairs that avoid $\mu$. The results (Ha $=H_a$, dHa $=H_a'$, and so on) are:

$$\begin{aligned}
E^{(1)0}{}_0 &= 3\left(H_a^2+4H_aH_b+2H_b^2\right),\\
E^{(1)1}{}_1 &= 2H_a'+4H_b'+3H_a^2+8H_aH_b+10H_b^2,\\
E^{(1)4}{}_4 &= 3H_a'+3H_b'+6H_a^2+9H_aH_b+6H_b^2.
\end{aligned}$$

$$
\begin{aligned}
E^{(2)0}{}_0 &= -12H_b\left(4H_a^3+18H_a^2H_b+12H_aH_b^2+H_b^3\right),\\
E^{(2)1}{}_1 &= -32H_a'H_aH_b-48H_a'H_b^2-16H_b'H_a^2-96H_b'H_aH_b\\
&\quad -48H_b'H_b^2-32H_a^3H_b-136H_a^2H_b^2-192H_aH_b^3\\
&\quad -60H_b^4,\\
E^{(2)4}{}_4 &= -12H_a'H_a^2-72H_a'H_aH_b-36H_a'H_b^2-36H_b'H_a^2\\
&\quad -72H_b'H_aH_b-12H_b'H_b^2-12H_a^4-108H_a^3H_b\\
&\quad -180H_a^2H_b^2-108H_aH_b^3-12H_b^4.
\end{aligned}
$$

$$
\begin{aligned}
E^{(3)0}{}_0 &= 360H_a^2H_b^3\left(4H_a+3H_b\right),\\
E^{(3)1}{}_1 &= 72H_b^2\left(8H_a'H_aH_b+2H_a'H_b^2+12H_b'H_a^2+8H_b'H_aH_b+8H_a^3H_b+19H_a^2H_b^2+8H_aH_b^3\right),\\
E^{(3)4}{}_4 &= 72H_aH_b\left(9H_a'H_aH_b+6H_a'H_b^2+6H_b'H_a^2+9H_b'H_aH_b+9H_a^3H_b+17H_a^2H_b^2+9H_aH_b^3\right).
\end{aligned}
$$

In particular

$$
E^{(2)0}{}_0 = -12H_b\big(4H_a^3+18H_a^2H_b+12H_aH_b^2+H_b^3\big),\qquad E^{(3)0}{}_0 = 360\,H_a^2H_b^3\big(4H_a+3H_b\big).
$$

Divergence-freeness becomes, for each $k$,

$$
\frac{d}{dx_0}E^{(k)0}{}_0+3H_a\big(E^{(k)0}{}_0-E^{(k)1}{}_1\big)+4H_b\big(E^{(k)0}{}_0-E^{(k)4}{}_4\big)=0,
$$

checked for arbitrary $a,b$. The VARIATIONAL ORIGIN is checked as well: with a lapse function $N(x_0)$ ($g_{00}=N^2$) the reduced Lagrangian is $Na^3b^4L_k$, and its Euler–Lagrange expressions at $N=1$ are

$$
\frac{\delta(Na^3b^4L_k)}{\delta N}=-2a^3b^4E^{(k)0}{}_0,\qquad \frac{\delta(Na^3b^4L_k)}{\delta a}=-6a^2b^4E^{(k)1}{}_1,\qquad \frac{\delta(Na^3b^4L_k)}{\delta b}=-8a^3b^3E^{(k)4}{}_4,
$$

for $k=1,2,3,4$ (for $k=4$ all three vanish identically: $\sqrt{|g|}L_4$ is a total derivative). The factors come from $\delta g^{00}=-2\delta N$, three $\delta g^{ii}=2\delta a/a^3$ and four $\delta g^{kk}=2\delta b/b^3$.

## 4.7 An inhomogeneous metric and a gravitational wave

For the explicit inhomogeneous metric $g=\mathrm{diag}(n^2,-a^2,-a^2,-a^2,-b^2,-b^2,-b^2,-b^2)$ with $n=1+x_0x_4/5+x_0^2/11$, $a=1+x_0/3+x_4^2/7$, $b=2-x_0x_4/4+x_0^2/9+x_4/13$, the divergence $\nabla_\mu E^{(k)\mu}{}_\nu$ is evaluated EXACTLY (rational arithmetic) at $x_0=1/2$, $x_4=1/3$ for $k=1,2,3$; it vanishes. The derivative of $E^{(k)}$ (a polynomial of degree $k$ in the curvature) is obtained from the curvature and its first derivatives at the point: $\partial_\lambda E^{(k)}=\frac{d}{d\epsilon}E^{(k)}[R+\epsilon\,\partial_\lambda R]\big|_{\epsilon=0}$. At that point $E^{(2)}$ equals the Lanczos tensor exactly, and

$$
E^{(3)0}{}_4\big|_{x_0=1/2,\,x_4=1/3}=q\approx 0.0143928243619,
$$

$q\in\mathbb{Q}$ (an exact fraction: numerator of 50, denominator of 52 digits).

For the pp-wave $ds^2=dx_0^2-dx_1^2-\cdots-dx_7^2+F(x_0-x_1,x_2,x_3)\,(dx_0-dx_1)^2$ with an arbitrary profile $F$ and the null vector $l_\mu=(1,-1,0,\ldots,0)$,

$$
E^{(1)}_{\mu\nu}=\tfrac12\big(\partial_2^2+\partial_3^2\big)F\;l_\mu l_\nu,\qquad E^{(2)}=E^{(3)}=0 \text{ for every profile } F,
$$

so a plane wave whose profile is harmonic in the transverse directions solves the Einstein–Lovelock equations for every $\alpha_2,\alpha_3$ (with $\bar\Lambda=0$). xAct and the pair-set sum agree for this metric, which is not diagonal.

## 4.8 Exponential vacuum solutions: eternal inflation of ordinary space and deflation of the auxiliary space

The procedure of the reference notebook — solve the vacuum equations of section 4.2 for $w_2$, $w_3$ (and $\Lambda$) — is applied to the cosmological metric with constant Hubble rates, $a=e^{H_ax_0}$, $b=e^{H_bx_0}$, $H_a=H$, $H_b=q_HH$ ($q_H$ a pure number; code `qH`). With $H_a'=H_b'=0$ in the closed forms of section 4.6 every $E^{(k)}$ is a constant

diagonal matrix proportional to $H^{2k}$, and the 64 equations reduce to three distinct ones (components 00, 11, 44). They are not independent: for constant Hubble rates the divergence identity of section 4.6 becomes

$$3H_a\big(\mathrm{eq}_{00}-\mathrm{eq}_{11}\big)+4H_b\big(\mathrm{eq}_{00}-\mathrm{eq}_{44}\big)=0,$$

so for every $(H_a,H_b)$ there is a one-parameter family of couplings: given $w_1$ and $\Lambda$ the equations fix $w_2$, $w_3$. For $q_H=-1$ — ordinary space expands as $e^{Hx_0}$ while the auxiliary space contracts as $e^{-Hx_0}$, for every time $x_0$ (eternal inflation of ordinary space with eternal deflation of the auxiliary space) — the three equations are

$$\mathrm{eq}_{00}=-\Lambda-3w_1-36w_2-360w_3,\qquad \mathrm{eq}_{11}=-\Lambda+5w_1+28w_2+216w_3,\qquad \mathrm{eq}_{44}=-\Lambda+3w_1+12w_2+72w_3,$$

and the family is

$$w_2=\frac{\Lambda}{4}-\frac{w_1}{2},\qquad w_3=-\frac{\Lambda}{36}+\frac{w_1}{24};$$
$$\Lambda=0:\quad \frac{w_2}{w_1}=-\frac{1}{2},\quad \frac{w_3}{w_1}=\frac{1}{24},\quad \text{i.e.}\quad \alpha_2=\frac{w_2}{w_1H^2}=-\frac{1}{2}\,H^{-2},\quad \alpha_3=\frac{w_3}{w_1H^4}=\frac{1}{24}\,H^{-4}.$$

For general $q_H$ (computed here from the closed forms of section 4.6; the notebook and the Python program obtain the same expressions, and check that they solve all 64 components for every $\Lambda$):

$$w_2=\frac{\big(12q_H^3+27q_H^2+22q_H+9\big)\,w_1-\big(2q_H+3\big)\,\Lambda}{4q_H\,\big(6q_H^4+36q_H^3+66q_H^2+51q_H+16\big)},$$

$$w_3=\frac{\big(15q_H^4+36q_H^3+33q_H^2+18q_H+3\big)\,w_1-\big(3q_H^2+6q_H+1\big)\,\Lambda}{72q_H^3\,\big(6q_H^4+36q_H^3+66q_H^2+51q_H+16\big)}.$$

# 5. Generalized frame matrices, canonical spin connection, total covariant derivative (section 13)

## 5.1 Frame field and generalized frame matrices

A frame field (vielbein) is a square root of the metric, and its inverse is written ${e_A}^\mu$:

$$g_{\mu\nu}=\eta_{AB}\,{e^A}_\mu {e^B}_\nu,\qquad {e_A}^\mu {e^A}_\nu=\delta^\mu_\nu.$$

The GENERALIZED FRAME MATRICES are the position-dependent real 16×16 matrices

$$\Gamma_\mu(X)={e^A}_\mu(X)\,\Gamma_A,\qquad \Gamma^\mu(X)={e_A}^\mu(X)\,\Gamma^A,\qquad \Gamma_\mu\Gamma_\nu+\Gamma_\nu\Gamma_\mu=2g_{\mu\nu}(X)\,1_{16},$$

with $\Gamma^A=\eta^{AA}\Gamma_A$ (no sum). They are real because the frame field and the $\Gamma_A$ are real.

## 5.2 The canonical spin connection and the total covariant derivative

The canonical (Levi-Civita) spin connection and its action on 16-component spinors are

$${\omega_\mu}^A{}_B={e^A}_\nu\big(\partial_\mu {e_B}^\nu+\{{}^\nu_{\mu\lambda}\}{e_B}^\lambda\big),\qquad \omega_{\mu AB}=\eta_{AC}\,{\omega_\mu}^C{}_B=-\omega_{\mu BA},$$
$$\Omega_\mu=\tfrac14\,\omega_{\mu AB}\,\Gamma^A\Gamma^B,\qquad D_\mu\psi=\partial_\mu\psi+\Omega_\mu\psi,\qquad D_\mu\bar\psi=\partial_\mu\bar\psi-\bar\psi\,\Omega_\mu.$$

The TOTAL covariant derivative acts on every index of an object: with the Christoffel symbols on coordinate indices, with $\omega$ on frame indices, and through the commutator with $\Omega$ on the two spinor indices of a 16×16 matrix. The canonical spin connection is the one for which the total covariant derivative of the frame field and of every frame matrix vanishes:

$$D_\mu {e^A}_\nu=\partial_\mu {e^A}_\nu-\{{}^\lambda_{\mu\nu}\}{e^A}_\lambda+{\omega_\mu}^A{}_B\,{e^B}_\nu=0\qquad\text{(vielbein postulate)},$$
$$\nabla_\mu\Gamma_\nu=\partial_\mu\Gamma_\nu-\{{}^\lambda_{\mu\nu}\}\Gamma_\lambda+[\Omega_\mu,\Gamma_\nu]=0,$$
$$D_\mu\Gamma^A={\omega_\mu}^A{}_B\,\Gamma^B+[\Omega_\mu,\Gamma^A]=0.$$

$\Omega_\mu$ is real and $\Gamma_0\Omega_\mu^{\mathsf T}\Gamma_0=-\Omega_\mu$, which makes $D_\mu\bar\psi$ the Dirac adjoint of $D_\mu\psi$ and the action below real.

## 5.3 Spin curvature, Bianchi identity, Lichnerowicz

The curvature of the spin connection is the Riemann tensor in spinor form:

$$\mathcal{F}_{\mu\nu} = \partial_\mu\Omega_\nu - \partial_\nu\Omega_\mu + [\Omega_\mu,\Omega_\nu] = [D_\mu,D_\nu] = \tfrac14\, R_{AB\mu\nu}\,\Gamma^A\Gamma^B, \qquad R_{AB\mu\nu} = \eta_{AC}\, e^C{}_\rho\, R^\rho{}_{\sigma\mu\nu}\, e_B{}^\sigma,$$
$$0 = \partial_\lambda\mathcal{F}_{\mu\nu} + [\Omega_\lambda,\mathcal{F}_{\mu\nu}] + \partial_\mu\mathcal{F}_{\nu\lambda} + [\Omega_\mu,\mathcal{F}_{\nu\lambda}] + \partial_\nu\mathcal{F}_{\lambda\mu} + [\Omega_\nu,\mathcal{F}_{\lambda\mu}] \qquad \text{(Bianchi)},$$
$$\tfrac12\,\Gamma^\mu\Gamma^\nu\mathcal{F}_{\mu\nu} = -\tfrac{R}{4}\,1_{16}, \qquad (\Gamma^\mu D_\mu)^2\psi = g^{\mu\nu}\big(\partial_\mu D_\nu\psi + \Omega_\mu D_\nu\psi - \{{}^{\lambda}_{\mu\nu}\}D_\lambda\psi\big) - \tfrac{R}{4}\psi.$$

Every identity of sections 5.1–5.3 is checked by the notebook (function `frameChecks`) and by the Python program for four metrics: the cosmological metric with the diagonal frame $e = \mathrm{diag}(1,a,a,a,b,b,b,b)$; the pp-wave with the non-diagonal frame $e^0 = (1+F/2)dx_0 - (F/2)dx_1$, $e^1 = (F/2)dx_0 + (1-F/2)dx_1$, $e^A = dx^A$ otherwise; the inhomogeneous metric with $e = \mathrm{diag}(n,a,a,a,b,b,b,b)$ (exactly at $x_0 = 1/2$, $x_4 = 1/3$, the derivatives being taken symbolically first); and the static radial metric of section 5.4.

## 5.4 Cartan's first structure equation; a static metric with a radial coordinate

With the frame one-forms $e^A = e^A{}_\mu dx^\mu$ and the connection one-forms $\omega^A{}_B = \omega_\mu{}^A{}_B\, dx^\mu$, the vielbein postulate antisymmetrized in $\mu,\nu$ (the Christoffel terms cancel) is Cartan's first structure equation without torsion:

$$de^A + \omega^A{}_B \wedge e^B = 0 \quad\Longleftrightarrow\quad \partial_\mu e^A{}_\nu - \partial_\nu e^A{}_\mu + \omega_\mu{}^A{}_B\, e^B{}_\nu - \omega_\nu{}^A{}_B\, e^B{}_\mu = 0.$$

It determines the canonical spin connection from the frame alone. With the structure coefficients $C^A{}_{BC} = -C^A{}_{CB}$ and the connection written with a frame form index,

$$de^A = \tfrac12\, C^A{}_{BC}\, e^B\wedge e^C, \qquad C^A{}_{BC} = e_B{}^\mu e_C{}^\nu\big(\partial_\mu e^A{}_\nu - \partial_\nu e^A{}_\mu\big), \qquad C_{ABC} = \eta_{AD}C^D{}_{BC},$$
$$\omega_{C,AB} \equiv e_C{}^\mu\,\omega_{\mu AB} = -\tfrac12\big(C_{ACB} + C_{BAC} - C_{CBA}\big),$$

the unique solution with $\omega_{C,AB} = -\omega_{C,BA}$; both relations are checked for all four frames. As an example take $r = x_1 > 0$ and arbitrary positive functions $f(r)$, $h(r)$ (code `fR[x1]`, `hR[x1]`):

$$ds^2 = f(r)\,dx_0^2 - h(r)\,dr^2 - r^2\big(dx_2^2 + dx_3^2 + dx_4^2 + dx_5^2 + dx_6^2 + dx_7^2\big),$$
$$e^0 = \sqrt{f}\,dx_0, \quad e^1 = \sqrt{h}\,dr, \quad e^k = r\,dx^k \ (k = 2,\dots,7).$$

The six directions $x_2,\dots,x_7$ are flat directions scaled by $r$ (a static "planar" metric of 7+1 spacetime, not a spherically symmetric one). Cartan's equation gives $de^0 = \frac{f'}{2\sqrt f}\,dr\wedge dx_0 = -\omega^0{}_1\wedge e^1$ and $de^k = dr\wedge dx^k = -\omega^k{}_1\wedge e^1$, hence $\omega^0{}_1 = \frac{f'}{2\sqrt{fh}}\,dx_0$, $\omega^k{}_1 = \frac{1}{\sqrt h}\,dx^k$, and

$$\Omega_0 = \frac{f'}{4\sqrt f\sqrt h}\,\Gamma_1\Gamma_0, \qquad \Omega_1 = 0, \qquad \Omega_k = \frac{1}{2\sqrt h}\,\Gamma_1\Gamma_k \quad (k = 2,\dots,7),$$
$$\Gamma^\mu\Omega_\mu = \frac{1}{\sqrt h}\Big(\frac{f'}{4f} + \frac{3}{r}\Big)\Gamma^1, \qquad \Gamma^1 = \eta^{11}\Gamma_1 = -\Gamma_1,$$

the $3/r = 6\times\frac{1}{2r}$ coming from the six scaled directions. The covariant Dirac operator and a rescaling that removes the connection are therefore

$$i\Gamma^\mu D_\mu - m = \frac{i}{\sqrt f}\,\Gamma^0\partial_0 + \frac{i}{\sqrt h}\,\Gamma^1\Big(\partial_r + \frac{f'}{4f} + \frac{3}{r}\Big) + \sum_{k=2}^{7}\frac{i}{r}\,\Gamma^k\partial_k - m,$$
$$\psi = f^{-1/4}r^{-3}\chi: \qquad \Gamma^\mu D_\mu\psi = f^{-1/4}r^{-3}\,\Gamma^\mu\partial_\mu\chi.$$

# 6. The covariant Dirac equation of dirac16complex

The Lagrangian density and the field equation are

$$\mathcal{L} = \sqrt{|g|}\;\bar\psi\big(i\,\Gamma^\mu D_\mu - m\big)\psi, \qquad \big(i\,\Gamma^\mu(X)\,D_\mu - m\big)\psi = 0,$$

where the spin connection $\Omega_\mu$ in $D_\mu$ is the coupling of gravity to dirac16complex. The current $J^\mu = \bar\psi\Gamma^\mu\psi$ is conserved because, for arbitrary $\psi$ and $\bar\psi$ (by $\nabla\Gamma = 0$ and $\partial_\mu\sqrt{|g|} = \sqrt{|g|}\{{}^{\lambda}_{\lambda\mu}\}$),

$$\partial_\mu\big(\sqrt{|g|}\,\bar\psi\Gamma^\mu\psi\big) = \sqrt{|g|}\,\big[(D_\mu\bar\psi)\Gamma^\mu\psi + \bar\psi\Gamma^\mu D_\mu\psi\big],$$

and the bracket vanishes when $\psi$ and $\bar\psi$ solve the field equation and its adjoint. Squaring with the Lichnerowicz formula gives $\big(g^{\mu\nu}\nabla_\mu D_\nu + m^2 - \frac{R}{4}\big)\psi = 0$.

**Coupling to a gauge field.** An abelian gauge field $A_\mu(X)$ with charge $q$ enters through the gauge-covariant derivative

$$D_\mu\psi = \partial_\mu\psi + \Omega_\mu\psi - iqA_\mu\psi, \qquad D_\mu\bar\psi = \partial_\mu\bar\psi - \bar\psi\,\Omega_\mu + iqA_\mu\bar\psi.$$

Under $\psi \to e^{iq\alpha}\psi$, $A_\mu \to A_\mu + \partial_\mu\alpha$ ($\alpha(X)$ arbitrary) the Dirac operator is covariant, $\Gamma^\mu D'_\mu(e^{iq\alpha}\psi) = e^{iq\alpha}\,\Gamma^\mu D_\mu\psi$, and the current identity above holds unchanged, so the electric current $q\,\bar\psi\Gamma^\mu\psi$ is conserved. Both statements are checked with the cosmological frame and arbitrary $A_\mu(x_0, x_1, x_5)$, $\alpha(x_0, x_1, x_5)$.

## 6.1 The cosmological spacetime

For the cosmological metric the spin connection is NOT zero whenever space expands:

$$\Omega_0 = 0, \qquad \Omega_i = \tfrac{a'}{2}\,\Gamma_i\Gamma_0\ (i = 1,2,3), \qquad \Omega_k = \tfrac{b'}{2}\,\Gamma_k\Gamma_0\ (k = 4,\dots,7), \qquad \Gamma^\mu\Omega_\mu = \Big(\tfrac{3}{2}H_a + 2H_b\Big)\Gamma_0.$$

The rescaled field $\chi = (a^3b^4)^{1/2}\psi$ obeys an equation without connection, $\Gamma^\mu D_\mu\psi = (a^3b^4)^{-1/2}\Gamma^\mu\partial_\mu\chi$. For a plane wave $\chi = e^{i(\vec p\cdot\vec x + \vec q\cdot\vec y)}c(x_0)$ — $\vec p = (p_1, p_2, p_3)$ the momentum along ordinary space $\vec x = (x_1, x_2, x_3)$, $\vec q = (q_4, \dots, q_7)$ along the auxiliary space $\vec y = (x_4, \dots, x_7)$ — the Dirac equation becomes

$$i\,\frac{dc}{dx_0} = h(x_0)\,c, \qquad h(x_0) = \sum_{i=1}^{3}\alpha_i\,\frac{p_i}{a} + \sum_{k=4}^{7}\alpha_k\,\frac{q_k}{b} + \beta\,m, \qquad \beta = \Gamma_0, \quad \alpha_j = \Gamma^0\Gamma^j = \begin{pmatrix} 0 & \tau_j^O \\ -\tau_j^O & 0 \end{pmatrix},$$

the flat-space Hamiltonian with the physical (redshifted) momenta $\vec p/a$ and $\vec q/b$.

## 6.2 Conformal weight 7/2

For a conformally flat metric $g = C(x_0)^2\eta$ ($x_0$ is then conformal time) the massless operator satisfies, for every $f$,

$$\Gamma^\mu D_\mu\big(C^{-7/2}f\big) = C^{-9/2}\,\Gamma^A\partial_A f,$$

so a massless dirac16complex in a conformally flat 7+1 spacetime is a rescaled flat-space solution and no pairs are created. The cosmological metric is conformally flat only when $a = b$.

## 6.3 Two classes of states in an anisotropic expansion

With unit vectors $\vec n_a$, $\vec n_b$ along $\vec p$ and $\vec q$ the Hamiltonian is $h = \frac{|\vec p|}{a}\mathcal{A} + \frac{|\vec q|}{b}\mathcal{B} + m\beta$ with the CONSTANT real matrices $\mathcal{A} = \sum_i n_{a,i}\alpha_i$, $\mathcal{B} = \sum_k n_{b,k}\alpha_k$. $\mathcal{A}$, $\mathcal{B}$, $\beta$ square to $1_{16}$ and anticommute; $\mathcal{Z} = \mathcal{A}\mathcal{B}\beta$ squares to $-1_{16}$, is antisymmetric, and commutes with $\mathcal{A}$, $\mathcal{B}$, $\beta$ and $h$, so its eigenvalue $\pm i$ is conserved. On $\mathcal{Z} = +i$ the three matrices act like the Pauli matrices $(\sigma_1, \sigma_2, \sigma_3)$, on $\mathcal{Z} = -i$ like $(\sigma_1, \sigma_2, -\sigma_3)$; the 16-component mode equation is four copies of each of

$$i\,\frac{dc}{dx_0} = \Big(\frac{|\vec p|}{a}\sigma_1 + \frac{|\vec q|}{b}\sigma_2 \pm m\,\sigma_3\Big)c, \qquad c \in \mathbb{C}^2.$$

Pairs are created when the direction of the physical momentum $(|\vec p|/a, |\vec q|/b)$ turns fast compared with the frequency $\omega = \sqrt{|\vec p|^2/a^2 + |\vec q|^2/b^2 + m^2}$. The Bogoliubov coefficient $|\beta|^2$ of a mode is the probability that a positive-energy state at the initial time ends in a negative-energy state at the final time; $|\alpha|^2 + |\beta|^2 = 1$.

# 7. The three epochs (section 14)

For the cosmological metric with $T^{\mu}{}_{\nu} = \mathrm{diag}(\varepsilon, -p_a, -p_a, -p_a, -p_b, -p_b, -p_b, -p_b)$ (energy density $\varepsilon$, pressures $p_a$ along ordinary space and $p_b$ along the auxiliary space) the field equations are

$$\sum_k \alpha_k E^{(k)0}{}_0 = \kappa\varepsilon, \qquad \sum_k \alpha_k E^{(k)1}{}_1 = -\kappa p_a, \qquad \sum_k \alpha_k E^{(k)4}{}_4 = -\kappa p_b,$$

and they imply the conservation law $\varepsilon' + 3H_a(\varepsilon + p_a) + 4H_b(\varepsilon + p_b) = 0$.

## 7.1 Primordial epoch (near $x_0 = 0$): Kasner solutions and dynamical compactification

For $a = x_0^p$, $b = x_0^q$ every term of $E^{(k)}$ scales as $x_0^{-2k}$: near $x_0 = 0$ the highest Lovelock order with a non-zero coupling dominates, and matter whose density grows more slowly than $x_0^{-2k}$ is negligible. With $H_a = p/x_0$, $H_b = q/x_0$, $H_a' = -p/x_0^2$, $H_b' = -q/x_0^2$ in the closed forms of section 4.6 the vacuum solutions of each order are:

- **Einstein** ($k = 1$): $3p + 4q = 1$ and $3p^2 + 4q^2 = 1$, i.e. $p = (1 \pm 2\sqrt{2})/7$, $q = (2 \mp 3\sqrt{2})/14$. On the branch $p = 0.5469$, $q = -0.1602$ ordinary space expands while the auxiliary space contracts as $x_0$ grows: a dynamical compactification of $x_4, \ldots, x_7$.
- **Gauss–Bonnet** ($k = 2$): $3p+4q = 3$, and $p$ is a root of $-203p^3+81p^2+351p+27 = 0$: $p = -1.0835018$, $-0.078631058$, $1.5611476$ (with $q = (3 - 3p)/4$).
- **Cubic Lovelock** ($k = 3$): the non-degenerate solution $(p, q) = (-15/7,\ 20/7)$ with $3p + 4q = 5$; $E^{(3)}$ vanishes identically on the degenerate families $q = 0$ (static auxiliary space) and $p = 0$.

The pure-order equations are (each multiplied by $x_0^{2k}$; columns 00, 11, 44):

$$\begin{aligned}
x_0^2 E^{(1)0}{}_0 &= 3\left(p^2 + 4pq + 2q^2\right), \\
x_0^2 E^{(1)1}{}_1 &= 3p^2 + 8pq - 2p + 10q^2 - 4q, \\
x_0^2 E^{(1)4}{}_4 &= 3\left(2p^2 + 3pq - p + 2q^2 - q\right).
\end{aligned}$$

$$\begin{aligned}
x_0^4 E^{(2)0}{}_0 &= -12q\left(4p^3 + 18p^2q + 12pq^2 + q^3\right), \\
x_0^4 E^{(2)1}{}_1 &= -4q\left(8p^3 + 34p^2q - 12p^2 + 48pq^2 - 36pq + 15q^3 - 12q^2\right), \\
x_0^4 E^{(2)4}{}_4 &= -12\left(p^4 + 9p^3q - p^3 + 15p^2q^2 - 9p^2q + 9pq^3 - 9pq^2 + q^4 - q^3\right).
\end{aligned}$$

$$\begin{aligned}
x_0^6 E^{(3)0}{}_0 &= 360p^2q^3\left(4p + 3q\right), \\
x_0^6 E^{(3)1}{}_1 &= 72pq^3\left(8p^2 + 19pq - 20p + 8q^2 - 10q\right), \\
x_0^6 E^{(3)4}{}_4 &= 72p^2q^2\left(9p^2 + 17pq - 15p + 9q^2 - 15q\right).
\end{aligned}$$

## 7.2 Pair creation in the primordial epoch

On the Einstein–Kasner branch a massless mode depends on $(|\vec{p}|, |\vec{q}|)$ only through the scaling variable $K = |\vec{p}|\,\lambda^{1-p}$, $\lambda = (|\vec{q}|/|\vec{p}|)^{1/(q-p)}$ ($x_0 \to \lambda x_0$ maps one problem onto the other). The evolution is adiabatic before and after the time at which $|\vec{p}|/a \approx |\vec{q}|/b$, so it is integrated from $10^{-15}\lambda$ to $300\lambda$:

| $(\lvert\vec{p}\rvert, \lvert\vec{q}\rvert)$ | $K$ | $\lvert\beta\rvert^2$ **(Mathematica)** | $\lvert\beta\rvert^2$ **(Python)** |
|---|---|---|---|
| $(1, 1)$ | 1.00000 | 0.010749346 | 0.010749346 |
| $(2, 1)$ | 3.11829 | 0.00013294108 | 0.00013294108 |
| $(1, 3)$ | 0.494632 | 0.067442022 | 0.067442022 |

The momenta $(1.4302\ldots, 2.5)$ have $K = 1$ and give the same $|\beta|^2$ as $(1, 1)$; an isotropic expansion creates no massless pairs ($|\beta|^2 < 10^{-20}$). With a mass the two classes $\mathcal{Z} = \pm i$ are created at different rates: for $m = 0.3$, $|\vec{p}| = 1$, $|\vec{q}| = 1.5$, $x_0$ from 0.05 to 40,

$$|\beta|^2_{16\times16} = 0.027052647 = \tfrac{1}{2}\big(|\beta|^2_{+m} + |\beta|^2_{-m}\big), \qquad |\beta|^2_{+m} = 0.025286593, \quad |\beta|^2_{-m} = 0.028818702.$$

## 7.3 Modern epoch: a static auxiliary space

With $b = b_0$ ($H_b = 0$) the Lovelock tensors reduce to (entries 00, 11, 44)

$$E^{(1)} = \mathrm{diag}\big(3H_a^2,\ 2H_a' + 3H_a^2,\ 3H_a' + 6H_a^2\big), \qquad E^{(2)} = \mathrm{diag}\big(0,\ 0,\ -12H_a^2(H_a' + H_a^2)\big), \qquad E^{(3)} = 0.$$

The 00 and 11 equations are the Friedmann equations of ordinary cosmology,

$$3H_a^2 = \kappa\varepsilon + \Lambda = 8\pi G_4\,\rho_4 + \Lambda, \qquad 2H_a' + 3H_a^2 = -\kappa p_a + \Lambda, \qquad G_4 = \frac{G_8}{V_\perp}, \quad V_\perp = b_0^4 V_c,$$

for matter confined to ordinary space ($\varepsilon = \rho_4/V_\perp$; $V_c$ the coordinate volume of the auxiliary space). The 44 equation then FIXES the stress that holds $b$ constant: $\kappa p_b = \Lambda - 3H_a' - 6H_a^2 + 12\alpha_2 H_a^2(H_a' + H_a^2)$ — the analogue of the fold fluid of Dirac_claude. A mode with auxiliary momentum $\vec{q}$ moves in ordinary space as a particle of mass $\sqrt{m^2 + |\vec{q}|^2/b_0^2}$ (a Kaluza–Klein tower on a torus). Pair creation today is suppressed by $e^{-2\pi M/H_0}$ (section 7.4) with $H_0 \approx 1.4\times10^{-42}$ GeV, i.e. absent.

## 7.4 Far future: de Sitter expansion and thermal pair creation

Eight-dimensional de Sitter space $a = b = e^{Hx_0}$ is an exact vacuum solution when all $E^{(k)}$ are proportional to $\delta$:

$$E^{(1)} = 21H^2\,\delta, \quad E^{(2)} = -420H^4\,\delta, \quad E^{(3)} = 2520H^6\,\delta \qquad \Longrightarrow \qquad 21\alpha_1 H^2 - 420\alpha_2 H^4 + 2520\alpha_3 H^6 = \Lambda.$$

Over a static auxiliary space ($a = e^{Hx_0}$, $b = b_0$): $E^{(1)} = \mathrm{diag}(3,3,6)H^2$, $E^{(2)} = \mathrm{diag}(0,0,-12)H^4$, $E^{(3)} = 0$. In both cases a mode of dirac16complex obeys $i\,dc/dx_0 = (k\,e^{-Hx_0}\sigma_1 + M\sigma_3)c$, with $M = m$, or with the Kaluza–Klein mass $M = \sqrt{m^2 + |\vec{q}|^2/b_0^2}$. Starting from the adiabatic vacuum at early times the number of created pairs per mode is exactly the Fermi–Dirac occupation at the Gibbons–Hawking temperature $T = H/(2\pi)$, independent of $k$:

$$|\beta|^2 = \frac{1}{e^{2\pi M/H} + 1}.$$

| $M/H$ | $\lvert\beta\rvert^2$ **(Mathematica)** | $1/(e^{2\pi M/H} + 1)$ |
|---|---|---|
| 0.25 | 0.172102919 | 0.172102899 |
| 0.5 | 0.0414238845 | 0.0414238322 |
| 1.0 | 0.00186397746 | 0.00186396189 |

The integrator is also checked on the Landau–Zener problem $i\,dc/dt = (\kappa\sigma_1 + t\sigma_3)c$, $t$ from $-150$ to 150, whose exact result is $|\beta|^2 = e^{-\pi\kappa^2}$.

# 8. Verification

Every statement above is a check of 7+1.nb (Wolfram Language 15.0.1) and of the independent Python program tools/ref_gravity.py (sympy, numpy, scipy), which rebuilds the Dirac matrices from 4+4.nb's split octonions and computes everything again with its own curvature and Lovelock engine (tools/lovelock71.py). The checks of sections 11–14 of the notebook:

| # | check | result | section |
|---|---|---|---|
| 1 | cosmological metric: sectional curvatures K_0i = -(Ha'+Ha^2), K_0k = -(Hb'+Hb^2), K_ij = -Ha^2, K_kl = -Hb^2, K_ik = -Ha Hb | True | 11.2 |
| 2 | cosmological metric: R^cd_ab is non-zero only for {c,d} = {a,b} | True | 11.2 |
| 3 | Einstein tensor: G^0_0 = 3Ha^2 + 12 Ha Hb + 6 Hb^2, G^1_1 = 2Ha' + 3Ha^2 + 4Hb' + 10Hb^2 + 8HaHb, G^4_4 = 3Ha' + 6Ha^2 + 3Hb' + 6Hb^2 + 9HaHb | True | 11.2 |
| 4 | scalar curvature R = -2 (3Ha' + 6Ha^2 + 4Hb' + 10Hb^2 + 12 Ha Hb) | True | 11.2 |
| 5 | inhomogeneous metric, exactly at (x0, x4) = (1/2, 1/3): R_rsmn = -R_srmn = -R_rsnm = R_mnrs and R^r_[smn] = 0 (first Bianchi identity) | True | 11.3 |
| 6 | inhomogeneous metric: contracted Bianchi identity nabla_mu G^mu_nu == 0, exactly at (x0, x4) = (1/2, 1/3) | True | 11.3 |
| 7 | cosmological metric: contracted Bianchi identity nabla_mu G^mu_nu = 0 (arbitrary a, b) | True | 11.3 |
| 8 | generalized Kronecker delta: contraction identity delta^{m1..mp r}_{n1..np r} = (8-p) delta^{m1..mp}_{n1..np} (p = 1..4, 60 random index sets each) | True | 12.1 |
| 9 | generalized Kronecker delta with 9 upper indices vanishes in 8 dimensions (100 random index sets) | True | 12.1 |
| 10 | xAct declaration as in the reference notebook: SignatureOfMetric[g71] == {7, 1, 0} (seven space-like dimensions, one time-like dimension, none degenerate) and SignDetOfMetric[g71] == -1 | True | 12.2 |
| 11 | eta17 is simply the squares of the octonion basis: eta17[[A+1, B+1]] == Re(e_A e_B), e_0^2 == +1 (time x0), e_i^2 == -1 (space x_i, i = 1..7) | True | 12.2 |
| 12 | the quadratic form of the octonions has the signs of the squares of the octonion basis: Re(x x) == x0^2 - x1^2 - ... - x7^2 == x.eta17.x for x = x^A e_A (symbolic) | True | 12.2 |
| 13 | xAct: Lovelock1 == R_a^b - (1/2) delta_a^b R, the Einstein tensor (abstract indices) | True | 12.3 |
| 14 | xAct: trace of Lovelock2 == -2 L_2 with L_2 = R^2 - 4 R_cd R^cd + R_cdfi R^cdfi (abstract indices) | True | 12.3 |
| 15 | xAct, cosmological metric: Lovelock1 == EinsteinCD == the Einstein tensor of the curvature engine of section 11.1 | True | 12.4 |
| 16 | pair-set algorithm == brute-force generalized-delta contraction for E^(1) (cosmological metric) | True | 12.5 |
| 17 | pair-set algorithm == xAct (section 12.4) for E^(1), E^(2), E^(3) of the cosmological metric (two independent computations) | True | 12.5 |
| 18 | E^(1) == Einstein tensor G and L_1 == R (cosmological metric) | True | 12.6 |
| 19 | trace identity E^(k)mu_mu == -((8-2k)/2) L_k for k = 0..4 | True | 12.6 |
| 20 | E^(4) == 0 identically in 8 dimensions, while the Euler density L_4 != 0 | True | 12.6 |
| 21 | every E^(k) is diagonal, E^(k)1_1 = E^(k)2_2 = E^(k)3_3 and E^(k)4_4 = ... = E^(k)7_7 | True | 12.6 |
| 22 | E^(2) == Lanczos tensor and L_2 == R^2 - 4 Ric.Ric + Riem.Riem (cosmological metric) | True | 12.6 |
| 23 | E^(2)0_0 = -12 Hb (4Ha^3 + 18Ha^2 Hb + 12 Ha Hb^2 + Hb^3) and E^(3)0_0 = 360 Ha^2 Hb^3 (4Ha + 3Hb) | True | 12.6 |
| 24 | divergence-free: d0 E^0_0 + 3Ha (E^0_0 - E^1_1) + 4Hb (E^0_0 - E^4_4) == 0 for k = 1, 2, 3 (arbitrary a, b) | True | 12.6 |
| 25 | variation of nL a^3 b^4 L_k: EL_nL = -2 a^3 b^4 E^(k)0_0, EL_a = -6 a^2 b^4 E^(k)1_1, EL_b = -8 a^3 b^3 E^(k)4_4 (k = 1..4) | True | 12.6 |
| 26 | inhomogeneous metric: nabla_mu E^(k)mu_nu == 0 EXACTLY at (x0, x4) = (1/2, 1/3) for k = 1, 2, 3 | True | 12.7 |
| 27 | inhomogeneous metric: E^(1) == G at the point, and E^(k)_mu nu symmetric (k = 1, 2, 3) | True | 12.7 |
| 28 | inhomogeneous metric: E^(2) == Lanczos tensor at the point (exact) | True | 12.7 |
| 29 | pp-wave: E^(1)_mu nu == +(1/2)(d2^2 + d3^2) fPP l_mu l_nu | True | 12.8 |
| 30 | pp-wave: E^(2) == E^(3) == 0 for every profile fPP | True | 12.8 |
| 31 | pp-wave: xAct (section 12.4) == pair-set algorithm for E^(1), E^(2), E^(3) (a metric that is not diagonal; arbitrary profile fPP) | True | 12.8 |
| 32 | exponential expansion a = e^{H x0}, b = e^{qH H x0}: xAct Lovelock_k == the closed forms of section 12.6 with Ha = H, Hb = qH H, Ha' = Hb' = 0 (k = 1, 2, 3) | True | 12.9 |
| 33 | exponential expansion: three distinct vacuum equations, related by 3 Ha (eq00 - eq11) + 4 Hb (eq00 - eq44) == 0 | True | 12.9 |

| # | check | result | section |
|---|---|---|---|
| 34 | exponential vacuum solutions: (w2, w3) solve all 64 components for every qH (generic qH) and every Lambda | True | 12.9 |
| 35 | eternal inflation (a = e^{H x0}) with deflation (b = e^{-H x0}): w2 == Lambda/4 - w1/2, w3 == w1/24 - Lambda/36; for Lambda = 0: w2/w1 == -1/2, w3/w1 == 1/24 | True | 12.9 |
| 36 | cosmological metric: Omega_0 == 0, Omega_i == (a'/2) Gamma17[i].Gamma17[0], Omega_k == (b'/2) Gamma17[k].Gamma17[0] | True | 13.2 |
| 37 | cosmological metric: Gamma^mu Omega_mu == (3a'/(2a) + 2b'/b) Gamma17[0] | True | 13.2 |
| 38 | current identity: d_mu(sqrt\|g\| psibar Gamma^mu psi) == sqrt\|g\| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi] (psi, psibar arbitrary) | True | 13.2 |
| 39 | Lichnerowicz formula: (Gamma^mu D_mu)^2 psi == (g^mu nu nabla_mu D_nu - R/4) psi | True | 13.2 |
| 40 | rescaling psi = (a^3 b^4)^(-1/2) chi removes the spin connection: Gamma^mu D_mu psi == (a^3 b^4)^(-1/2) Gamma^mu d_mu chi | True | 13.2 |
| 41 | mode equation: Gamma17[0].(i Gamma^mu D_mu - m) psi == (a^3 b^4)^(-1/2) e^{i(p.x+q.y)} (i dc/dx0 - h(x0) c) for psi = (a^3 b^4)^(-1/2) e^{i(p.x+q.y)} c(x0) | True | 13.2 |
| 42 | conformal weight 7/2: Gamma^mu D_mu (C^(-7/2) f) == C^(-9/2) Gamma17up[A] d_A f for g = C(x0)^2 eta17 | True | 13.4 |
| 43 | Cartan's first structure equation de^A + omega^A_B ^ e^B == 0 (no torsion): cosmological, pp-wave and inhomogeneous frames | True | 13.5 |
| 44 | structure coefficients: omega_C,AB == -(1/2)(C_ACB + C_BAC - C_CBA) with C^A_BC = e_B^mu e_C^nu (d_mu e^A_nu - d_nu e^A_mu): cosmological and inhomogeneous frames | True | 13.5 |
| 45 | static radial metric: Cartan's first structure equation and omega_C,AB == -(1/2)(C_ACB + C_BAC - C_CBA) | True | 13.5 |
| 46 | static radial metric: Omega_0 == (f'/(4 Sqrt[f] Sqrt[h])) Gamma17[1].Gamma17[0], Omega_1 == 0, Omega_k == (1/(2 Sqrt[h])) Gamma17[1].Gamma17[k] (k = 2..7) | True | 13.5 |
| 47 | static radial metric: Gamma^mu Omega_mu == (1/Sqrt[h]) (f'/(4 f) + 3/r) Gamma17up[1] | True | 13.5 |
| 48 | static radial metric: psi = f^(-1/4) r^(-3) chi removes the spin connection, Gamma^mu D_mu psi == f^(-1/4) r^(-3) Gamma^mu d_mu chi | True | 13.5 |
| 49 | U(1) gauge covariance: Gamma^mu (D_mu - i q A'_mu)(e^{i q alpha} psi) == e^{i q alpha} Gamma^mu (D_mu - i q A_mu) psi with A'_mu = A_mu + d_mu alpha | True | 13.5 |
| 50 | current identity with the gauge field: d_mu(sqrt\|g\| psibar Gamma^mu psi) == sqrt\|g\| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi], D including -i q A_mu | True | 13.5 |
| 51 | Kasner metric: the Einstein tensor computed directly from the metric equals the closed form (x0^2 G^mu_mu, mu = 0, 1, 4) | True | 14.1 |
| 52 | Einstein Kasner: (p, q) = ((1+2Sqrt[2])/7, (2-3Sqrt[2])/14) and ((1-2Sqrt[2])/7, (2+3Sqrt[2])/14) solve E^(1) == 0 | True | 14.1 |
| 53 | Gauss-Bonnet Kasner: 3 real roots of the cubic; (p, (3-3p)/4) solve E^(2) == 0 | True | 14.1 |
| 54 | cubic Lovelock Kasner: (p, q) = (-15/7, 20/7) solves E^(3) == 0; E^(3) == 0 identically for q = 0 and for p = 0 | True | 14.1 |
| 55 | 2x2 reduction: A, B, betaD square to ID16 and anticommute; Zc = A.B.betaD: Zc.Zc == -ID16, Transpose[Zc] == -Zc, Zc commutes with A, B, betaD | True | 14.2 |
| 56 | Kasner pair creation: \|alpha\|^2 + \|beta\|^2 == 1 (to 10^-8) for (\|p\|, \|q\|) = (1,1), (2,1), (1,3) | True | 14.3 |
| 57 | Kasner scaling: (\|p\|, \|q\|) = (1.4302.., 2.5) has K = 1 and the same \|beta\|^2 as (1, 1) (to 10^-6) | True | 14.3 |
| 58 | 16x16 evolution == 4 copies of the 2x2 evolution with +m sigma3 and 4 with -m sigma3 (Kasner, m = 0.3, \|p\| = 1, \|q\| = 1.5, x0 from 0.05 to 40): average \|beta\|^2 agrees to 10^-8; unitarity; the two classes differ | True | 14.3 |
| 59 | isotropic expansion, massless: no pair creation (\|beta\|^2 < 10^-20) | True | 14.3 |
| 60 | static auxiliary space (Hb = 0): E^(1) = (3Ha^2, 2Ha' + 3Ha^2, 3Ha' + 6Ha^2), E^(2) = (0, 0, -12 Ha^2 (Ha' + Ha^2)), E^(3) = 0 | True | 14.4 |
| 61 | 8D de Sitter: E^(1) = 21 H^2, E^(2) = -420 H^4, E^(3) = 2520 H^6 (all diagonal entries) | True | 14.5 |
| 62 | de Sitter over a static auxiliary space: E^(1) = (3, 3, 6) H^2, E^(2) = (0, 0, -12) H^4, E^(3) = 0 | True | 14.5 |
| 63 | de Sitter pair creation: \|beta\|^2 == 1/(Exp[2 Pi M/H] + 1) to 10^-7, M/H = 0.25, 0.5, 1 | True | 14.5 |
| 64 | integrator benchmark (Landau-Zener): \|beta\|^2 == Exp[-Pi kappa^2] to 10^-6 | True | 14.5 |
| 65 | cosmological metric: eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real) | True | 13.1 |
| 66 | cosmological metric: canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA | True | 13.1 |
| 67 | cosmological metric: vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0 | True | 13.1 |
| 68 | cosmological metric: TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0 | True | 13.1 |
| 69 | cosmological metric: constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0 | True | 13.1 |

| # | check | result | section |
|---|---|---|---|
| 70 | cosmological metric: Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu) | True | 13.1 |
| 71 | cosmological metric: spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B] | True | 13.1 |
| 72 | cosmological metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0 | True | 13.1 |
| 73 | cosmological metric: (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16 | True | 13.1 |
| 74 | pp-wave: eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real) | True | 13.1 |
| 75 | pp-wave: canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA | True | 13.1 |
| 76 | pp-wave: vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0 | True | 13.1 |
| 77 | pp-wave: TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0 | True | 13.1 |
| 78 | pp-wave: constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0 | True | 13.1 |
| 79 | pp-wave: Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu) | True | 13.1 |
| 80 | pp-wave: spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B] | True | 13.1 |
| 81 | pp-wave: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0 | True | 13.1 |
| 82 | pp-wave: (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16 | True | 13.1 |
| 83 | inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real) | True | 13.1 |
| 84 | inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA | True | 13.1 |
| 85 | inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0 | True | 13.1 |
| 86 | inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0 | True | 13.1 |
| 87 | inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0 | True | 13.1 |
| 88 | inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu) | True | 13.1 |
| 89 | inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B] | True | 13.1 |
| 90 | inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0 | True | 13.1 |
| 91 | inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16 | True | 13.1 |
| 92 | static radial metric: eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real) | True | 13.1 |
| 93 | static radial metric: canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA | True | 13.1 |
| 94 | static radial metric: vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0 | True | 13.1 |
| 95 | static radial metric: TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0 | True | 13.1 |
| 96 | static radial metric: constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0 | True | 13.1 |
| 97 | static radial metric: Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu) | True | 13.1 |
| 98 | static radial metric: spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B] | True | 13.1 |
| 99 | static radial metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0 | True | 13.1 |
| 100 | static radial metric: (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16 | True | 13.1 |

The Python program's boolean checks:

| # | check | result |
|---|---|---|
| 1 | eta17 is simply the squares of the octonion basis: eta_AB = Re(e_A e_B), e_0^2 = +1, e_i^2 = -1 (i = 1..7) | True |
| 2 | the quadratic form of the octonions has the signs of the squares of the octonion basis: Re(x x) = x0^2 - x1^2 - ... - x7^2 = x.eta17.x | True |
| 3 | cosmological metric: sectional curvatures R^{AB}_{AB} = -(Ha'+Ha^2), -(Hb'+Hb^2), -Ha^2, -Hb^2, -Ha Hb | True |

| # | check | result |
|---|---|---|
| 4 | cosmological metric: Riemann R^{cd}_{ab} non-zero only for {c,d} = {a,b} | True |
| 5 | G^0_0 = 3Ha^2 + 12 Ha Hb + 6 Hb^2 | True |
| 6 | G^1_1 = 2Ha' + 3Ha^2 + 4Hb' + 10Hb^2 + 8HaHb | True |
| 7 | G^4_4 = 3Ha' + 6Ha^2 + 3Hb' + 6Hb^2 + 9HaHb | True |
| 8 | scalar curvature R = -2(3Ha' + 6Ha^2 + 4Hb' + 10Hb^2 + 12 HaHb) | True |
| 9 | E^(1) = Einstein tensor G (cosmological metric) | True |
| 10 | L_1 = R | True |
| 11 | E^(0) = -delta/2 | True |
| 12 | trace identity E^(k)mu_mu = -(8-2k)/2 L_k, k = 0..4 | True |
| 13 | E^(4) = 0 identically in 8 dimensions, but L_4 (Euler density) is not 0 | True |
| 14 | E^(k) diagonal: E^(k)mu_nu = 0 for mu != nu, and E^(k)1_1 = E^(k)2_2 = E^(k)3_3, E^(k)4_4 = ... = E^(k)7_7 | True |
| 15 | E^(2)0_0 = -12 Hb (4Ha^3 + 18Ha^2Hb + 12HaHb^2 + Hb^3) | True |
| 16 | E^(3)0_0 = 360 Ha^2 Hb^3 (4Ha + 3Hb) | True |
| 17 | E^(2) = Lanczos tensor 2(R R_mn - 2R_ma R^a_n - 2R^ab R_manb + R_m^abc R_nabc) - g_mn L_2/2, and L_2 = R^2 - 4 Ric.Ric + Riem.Riem (cosmological metric) | True |
| 18 | divergence-free E^(k), k = 1, 2, 3 (cosmological metric, arbitrary a, b) | True |
| 19 | variation of sqrt\|g\| L_k (lapse N, a, b): EL_N = -2 a^3 b^4 E^(k)0_0, EL_a = -6 a^2 b^4 E^(k)1_1, EL_b = -8 a^3 b^3 E^(k)4_4, k = 1..4 | True |
| 20 | inhomogeneous metric (N, A, B of x0 and x4): E^(1) = G at the point | True |
| 21 | inhomogeneous metric: nabla_mu E^(k)mu_nu = 0 exactly at x0 = 1/2, x4 = 1/3, k = 1, 2, 3 | True |
| 22 | inhomogeneous metric: E^(k)_mu nu symmetric, k = 1, 2, 3 | True |
| 23 | pp-wave: E^(1)_mu nu = +(1/2) (d2^2 + d3^2) F l_mu l_nu, l = dx0 - dx1 | True |
| 24 | pp-wave: E^(2) = E^(3) = 0 for every profile F | True |
| 25 | Einstein Kasner: p = (1+-2 sqrt2)/7, q = (2-+3 sqrt2)/14 solve E^(1) = 0; 3p+4q = 1, 3p^2+4q^2 = 1 | True |
| 26 | Gauss-Bonnet Kasner: the three (p, (3-3p)/4) solve E^(2) = 0, sum of the 7 exponents = 3 | True |
| 27 | cubic Lovelock Kasner: (p, q) = (-15/7, 20/7) solves E^(3) = 0, 3p + 4q = 5; degenerate families q = 0 and p = 0 | True |
| 28 | Einstein Kasner is not a cubic-Lovelock vacuum (E^(3) != 0) | True |
| 29 | 8D de Sitter (a = b = e^{H x0}): E^(1) = 21 H^2, E^(2) = -420 H^4, E^(3) = 2520 H^6 (times delta) | True |
| 30 | 4D de Sitter on a static T^4 (a = e^{H x0}, b = b0): E^(1) = diag(3,3,6) H^2, E^(2) = diag(0,0,-12) H^4, E^(3) = 0 | True |
| 31 | static T^4 (Hb = 0): E^(1) = (3Ha^2, 2Ha'+3Ha^2, 3Ha'+6Ha^2), E^(2) = (0, 0, -12 Ha^2 (Ha'+Ha^2)), E^(3) = 0 | True |
| 32 | exponential expansion a = e^{H x0}, b = e^{q H x0}: E^(k) (00, 11, 44) = closed forms with Ha = H, Hb = q H, Ha' = Hb' = 0 (k = 1, 2, 3) | True |
| 33 | exponential expansion: 3 Ha (eq00 - eq11) + 4 Hb (eq00 - eq44) = 0 identically (the three vacuum equations have rank 2) | True |
| 34 | exponential vacuum solutions: one solution (w2, w3) for given w1, Lambda, q, satisfying all three equations | True |
| 35 | eternal inflation a = e^{H x0} with deflation b = e^{-H x0}: w2 = Lambda/4 - w1/2, w3 = w1/24 - Lambda/36 (Lambda = 0: w2/w1 = -1/2, w3/w1 = 1/24) | True |
| 36 | cosmological metric: eta e e = g | True |
| 37 | cosmological metric: {Gamma_mu, Gamma_nu} = 2 g_mu nu | True |
| 38 | cosmological metric: Gamma_mu real | True |
| 39 | cosmological metric: omega_mu AB = -omega_mu BA | True |
| 40 | cosmological metric: vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0 | True |
| 41 | cosmological metric: Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu | True |
| 42 | cosmological metric: total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0 | True |
| 43 | cosmological metric: constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0 | True |
| 44 | cosmological metric: spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B | True |
| 45 | cosmological metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0 | True |
| 46 | cosmological metric: (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16 | True |
| 47 | cosmological metric: Gamma^mu Omega_mu = (3a'/(2a) + 2b'/b) Gamma17[0] | True |
| 48 | cosmological metric: Omega_i = (a'/2) Gamma17[i] Gamma17[0] (i = 1..3), Omega_k = (b'/2) Gamma17[k] Gamma17[0] (k = 4..7), Omega_0 = 0 | True |
| 49 | pp-wave: eta e e = g | True |
| 50 | pp-wave: {Gamma_mu, Gamma_nu} = 2 g_mu nu | True |
| 51 | pp-wave: Gamma_mu real | True |
| 52 | pp-wave: omega_mu AB = -omega_mu BA | True |
| 53 | pp-wave: vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0 | True |
| 54 | pp-wave: Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu | True |

| # | check | result |
|---|---|---|
| 55 | pp-wave: total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0 | True |
| 56 | pp-wave: constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0 | True |
| 57 | pp-wave: spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B | True |
| 58 | pp-wave: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0 | True |
| 59 | pp-wave: (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16 | True |
| 60 | inhomogeneous metric N, A, B (x0, x4): eta e e = g | True |
| 61 | inhomogeneous metric N, A, B (x0, x4): {Gamma_mu, Gamma_nu} = 2 g_mu nu | True |
| 62 | inhomogeneous metric N, A, B (x0, x4): Gamma_mu real | True |
| 63 | inhomogeneous metric N, A, B (x0, x4): omega_mu AB = -omega_mu BA | True |
| 64 | inhomogeneous metric N, A, B (x0, x4): vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0 | True |
| 65 | inhomogeneous metric N, A, B (x0, x4): Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu | True |
| 66 | inhomogeneous metric N, A, B (x0, x4): total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0 | True |
| 67 | inhomogeneous metric N, A, B (x0, x4): constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0 | True |
| 68 | inhomogeneous metric N, A, B (x0, x4): spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B | True |
| 69 | inhomogeneous metric N, A, B (x0, x4): Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0 | True |
| 70 | inhomogeneous metric N, A, B (x0, x4): (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16 | True |
| 71 | Cartan's first structure equation de^A + omega^A_B ^ e^B = 0 (torsion-free) for the cosmological frame | True |
| 72 | static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): eta e e = g | True |
| 73 | static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): {Gamma_mu, Gamma_nu} = 2 g_mu nu | True |
| 74 | static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Gamma_mu real | True |
| 75 | static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): omega_mu AB = -omega_mu BA | True |
| 76 | static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0 | True |
| 77 | static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu | True |
| 78 | static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0 | True |
| 79 | static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0 | True |
| 80 | static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B | True |
| 81 | static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0 | True |
| 82 | static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16 | True |
| 83 | Cartan's first structure equation for the static radial metric | True |
| 84 | structure coefficients: de^A = (1/2) C^A_BC e^B ^ e^C and omega_C,AB (form index C) = -(1/2)(C_ACB + C_BAC - C_CBA) (static radial metric) | True |
| 85 | static radial metric: Omega_0 = (f'/(4 sqrt(f h))) Gamma17[1] Gamma17[0], Omega_1 = 0, Omega_k = (1/(2 sqrt h)) Gamma17[1] Gamma17[k] (k = 2..7) | True |
| 86 | static radial metric: Gamma^mu Omega_mu = (1/sqrt h)(f'/(4f) + 3/r) Gamma17up[1], so (i Gamma^mu D_mu - m) psi = [i Gamma17up[0] d0/sqrt f + i Gamma17up[1] (d_r + f'/(4f) + 3/r)/sqrt h + sum_k i Gamma17up[k] d_k/r - m] psi | True |
| 87 | static radial metric: psi = f^(-1/4) r^(-3) chi removes the spin connection: Gamma^mu D_mu psi = f^(-1/4) r^(-3) Gamma^mu d_mu chi | True |
| 88 | U(1) gauge covariance: Gamma^mu (D_mu - i q A'_mu)(e^{i q alpha} psi) = e^{i q alpha} Gamma^mu (D_mu - i q A_mu) psi with A' = A + d alpha | True |
| 89 | current identity with the gauge field: d_mu(sqrt\|g\| psibar Gamma^mu psi) = sqrt\|g\| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi], D including -i q A_mu | True |
| 90 | current identity d_mu(sqrt\|g\| psibar Gamma^mu psi) = sqrt\|g\| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi] | True |
| 91 | Lichnerowicz: (Gamma^mu D_mu)^2 psi = (g^mu nu nabla_mu D_nu - R/4) psi | True |
| 92 | rescaling psi = (a^3 b^4)^(-1/2) chi removes the spin connection: Gamma^mu D_mu psi = (a^3 b^4)^(-1/2) Gamma^mu d_mu chi | True |
| 93 | mode equation: (i Gamma^mu D_mu - m) psi = 0 <=> i dc/dx0 = h(x0) c, h = alpha.(p/a, q/b) + beta m | True |

| # | check | result |
|---|---|---|
| 94 | conformal weight 7/2: Gamma^mu D_mu (C^(-7/2) f) = C^(-9/2) Gamma^A d_A f for g = C(x0)^2 eta | True |
| 95 | 2x2 reduction: A = alpha.n_a, B = alpha.n_b, beta: squares 1, mutually anticommuting; Z = A B beta: Z^2 = -1, Z^T = -Z, commutes with A, B, beta | True |
| 96 | Kasner: \|alpha\|^2 + \|beta\|^2 = 1 (to 1e-8) | True |
| 97 | Kasner scaling: (\|p\|, \|q\|) = (1.430..., 2.5) has K = 1 and the same \|beta\|^2 as (1, 1) (to 1e-6) | True |
| 98 | 16x16 evolution = 4 copies of the 2x2 evolution with +m sigma3 and 4 copies with -m sigma3: average agrees (to 1e-8), unitarity; the two sectors differ | True |
| 99 | isotropic expansion, massless: no pair creation (\|beta\|^2 < 1e-20) | True |
| 100 | de Sitter: \|beta\|^2 = 1/(exp(2 pi m/H) + 1) (Fermi-Dirac at the Gibbons-Hawking temperature H/2pi), to 1e-7 | True |
| 101 | integrator benchmark: Landau-Zener \|beta\|^2 = exp(-pi kappa^2) (to 1e-6) | True |

# 9. How to reproduce

1. `python3 tools/make_7p1_nb.py` writes 7+1.nb (unevaluated) from tools/make_7p1_nb.py and tools/nb_part_*.py.
2. Evaluate 7+1.nb in a fresh kernel of Wolfram 15.0.1 (Evaluation ▷ Quit Kernel, then Evaluation ▷ Evaluate Notebook; or `Get["D:/Developer/github/7+1/tools/run_eval.wl"]`) and save it; the last section summarizes all checks.
3. `python3 tools/ref_gravity.py` repeats the checks of this document independently (exit code 0 if all pass).
4. `python3 tools/make_docs.py` regenerates this document, its symbol dictionary and its provenance record.